AI LAB

1.3&4 – LET'S LOOK AT SOME DATA FOR REAL

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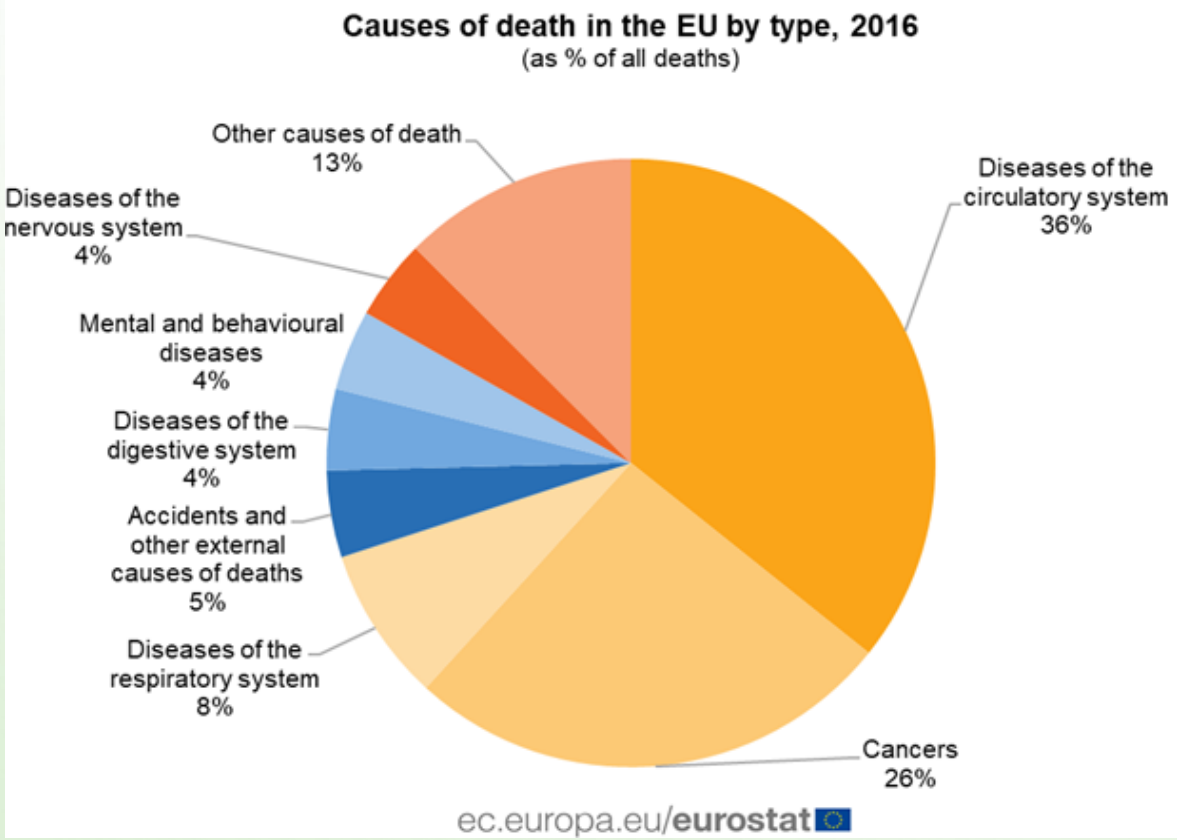
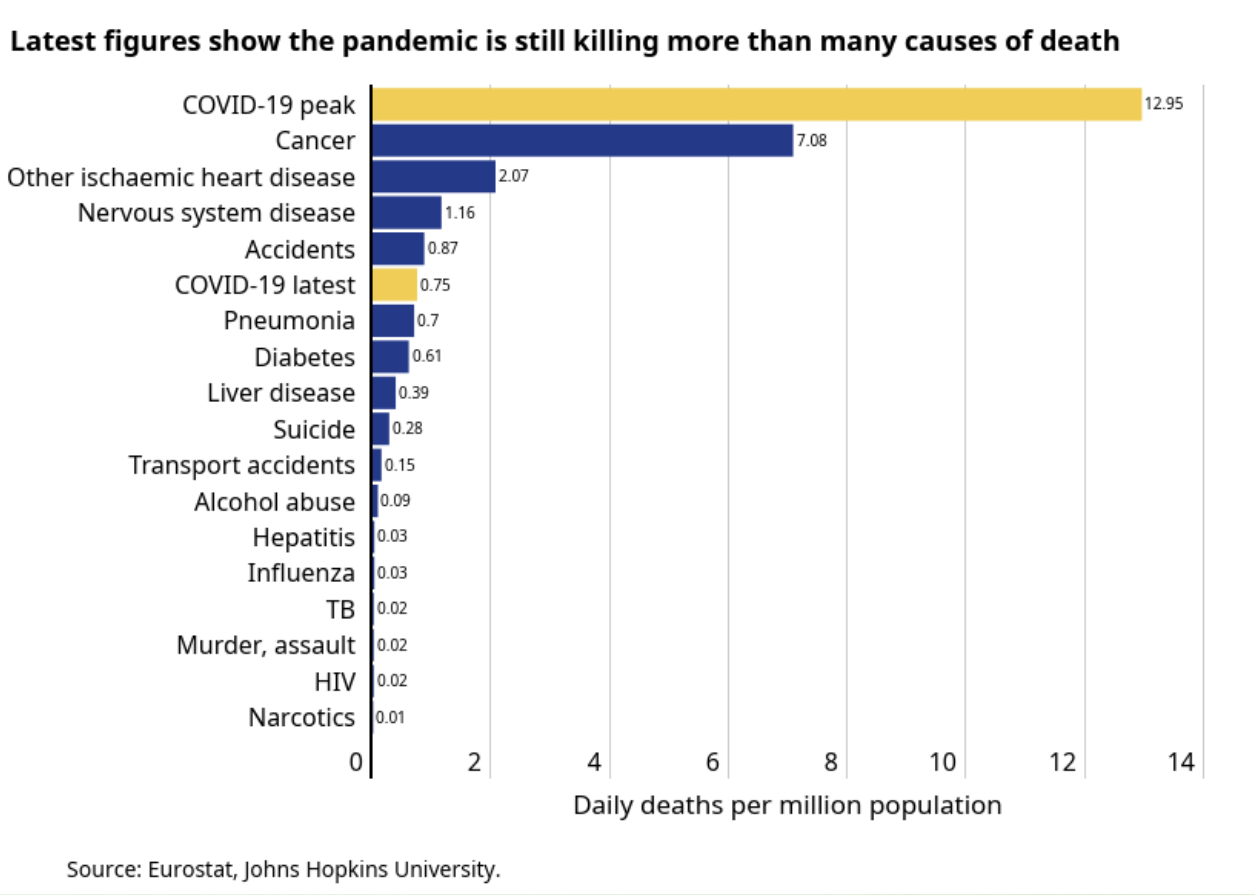
CHALLENGE - PART 1

- DIVIDE INTO GROUPS, 4-5 PEOPLE PER GROUP IF POSSIBLE
- DOWNLOAD 3 GENE EXPRESSION MATRIXES FROM LINK-D
- <u>THE DATA YOU WILL DOWNLOAD ARE CONFIDENTIAL AND FOR USE IN THIS COURSE ONLY</u>
- LOOK AT THE DATA DISTRIBUTION (PLOT, CALCULATE DESCRIPTIVE STATISTICS, ETC)
- LOOK AT THE DATA STRUCTURE (PCA, CLUSTERING)
- COMPARE THE 3 DATASETS
- WRITE REPORTS WITH YOUR INITIAL COMPARISON OF THE 3 DATASETS,
- INCLUDE THE CODE YOU USED TO GENERATE THE REPORTS
- UPLOAD YOUR REPORTS TO LINK-U

SOME BACKGROUND ON THE DATA YOU RECEIVED: CANCER

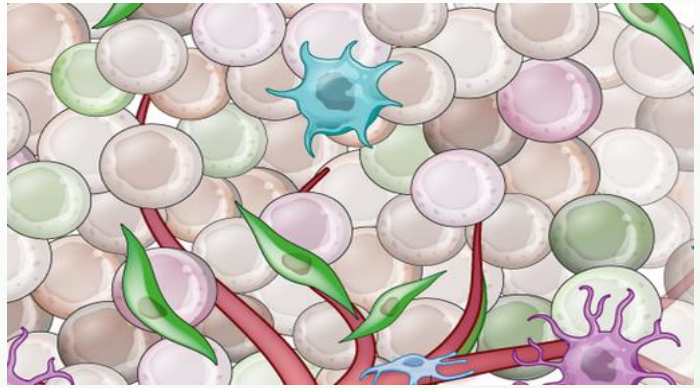




THE CHALLENGE

Cancer is a complex and heterogeneous disease

1. genetics
2. cell-cell signalling
3. microenvironment



Many FDA approved drugs available

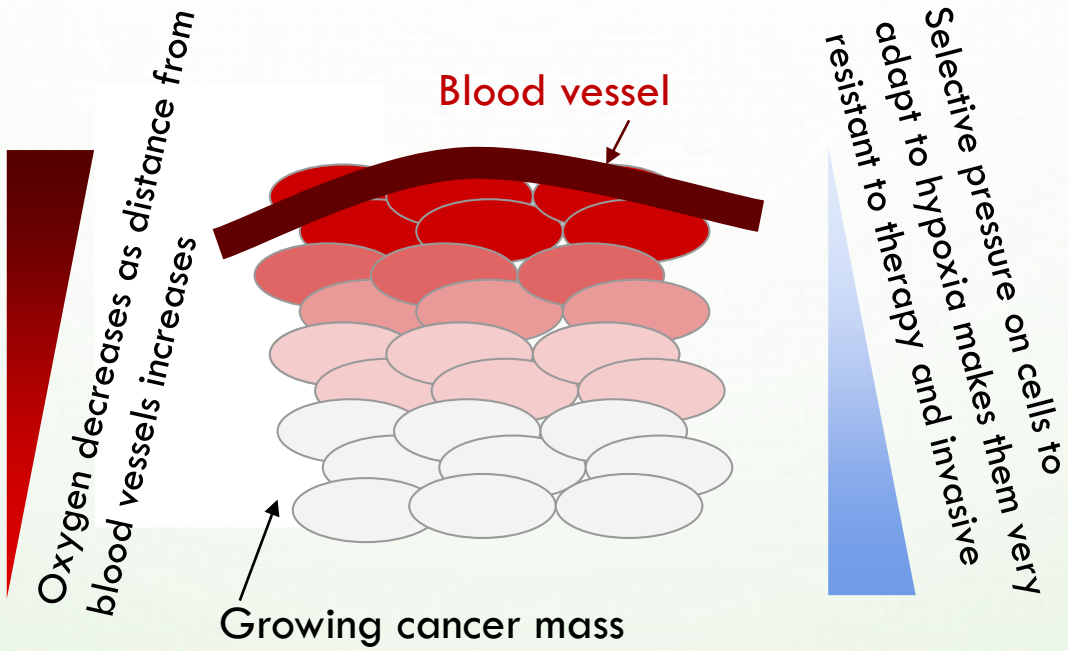
Cetuximab	*EGFR*
Bevacizumab	*VEGF*
Olaparib	*PARP*
Enzalutamide	*AR*
Pictilisib	*PI3K*

→ Therapeutic resistance

Hypoxia in cancer: a cause of therapeutic resistance and linked with poor prognosis

Hypoxia key trait of cancer with respect to normal tissues
Genome Med (2011)

Key methodology to identify hypoxic tumours
Cancer Cell (2013)

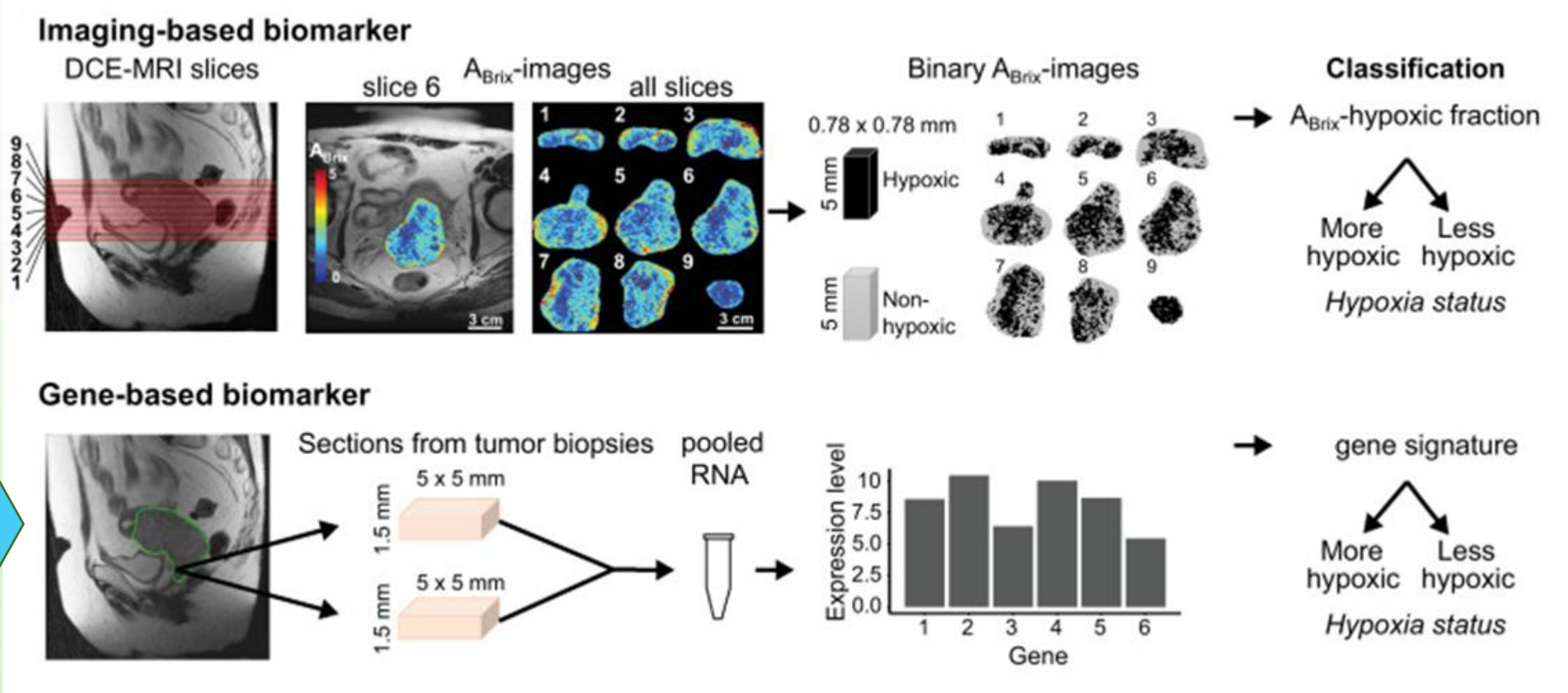


Hypoxia, poor prognosis and treatment resistance
JNCI (2011)

Hypoxia, genomic instability and selective pressure
Genome Biology (2016)

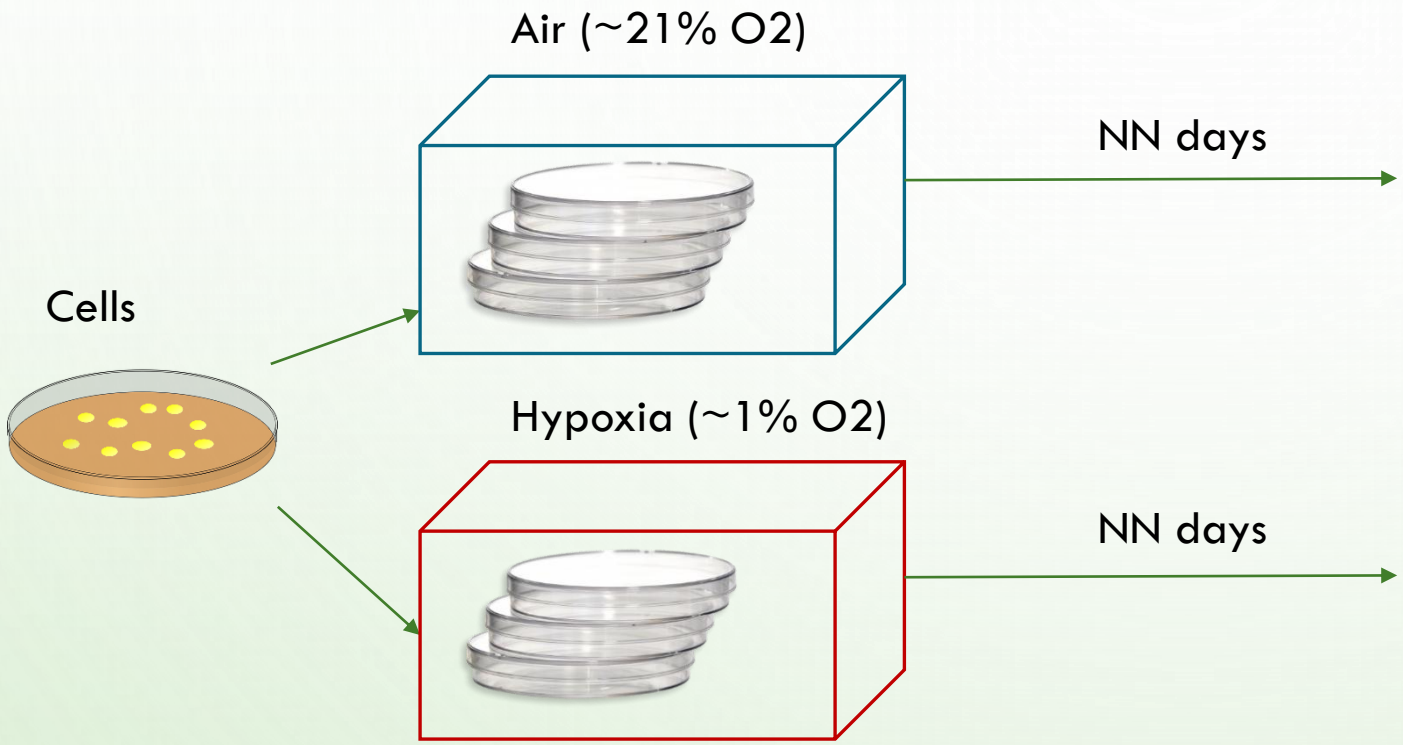
Can we identify hypoxic cells?

HOW DO WE MEASURE HYPOXIA IN CANCER?



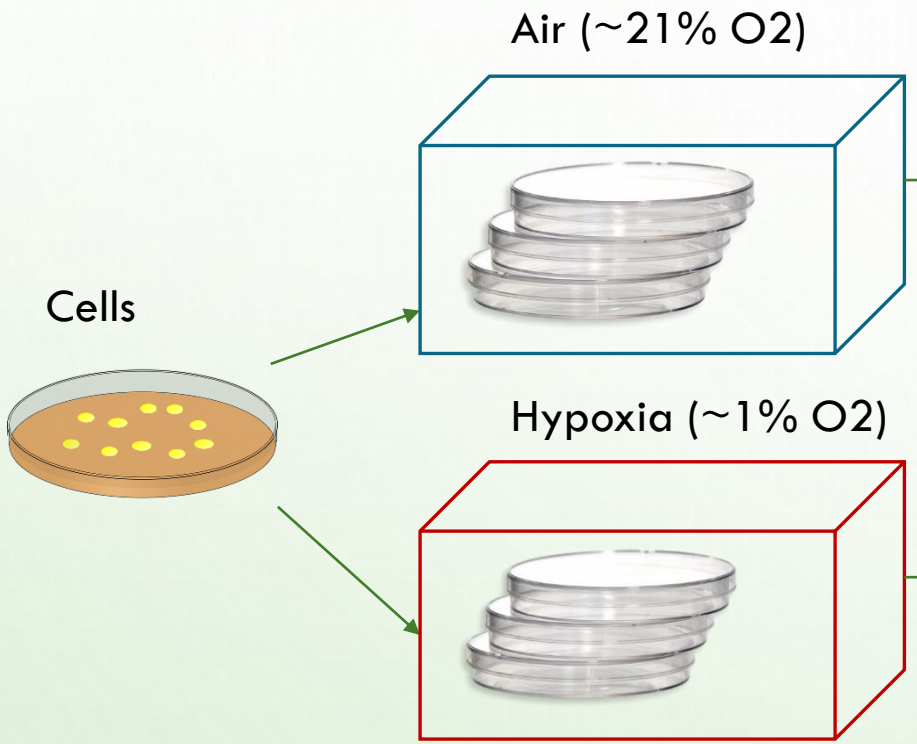
Adapted from https://www.thelancet.com/journals/ebiom/article/PIIS2352-3964(20)30216-4/fulltext

DERIVE GENE BASED BIOMARKER IN THE LAB



Sequencing/Omics
Imaging
Others

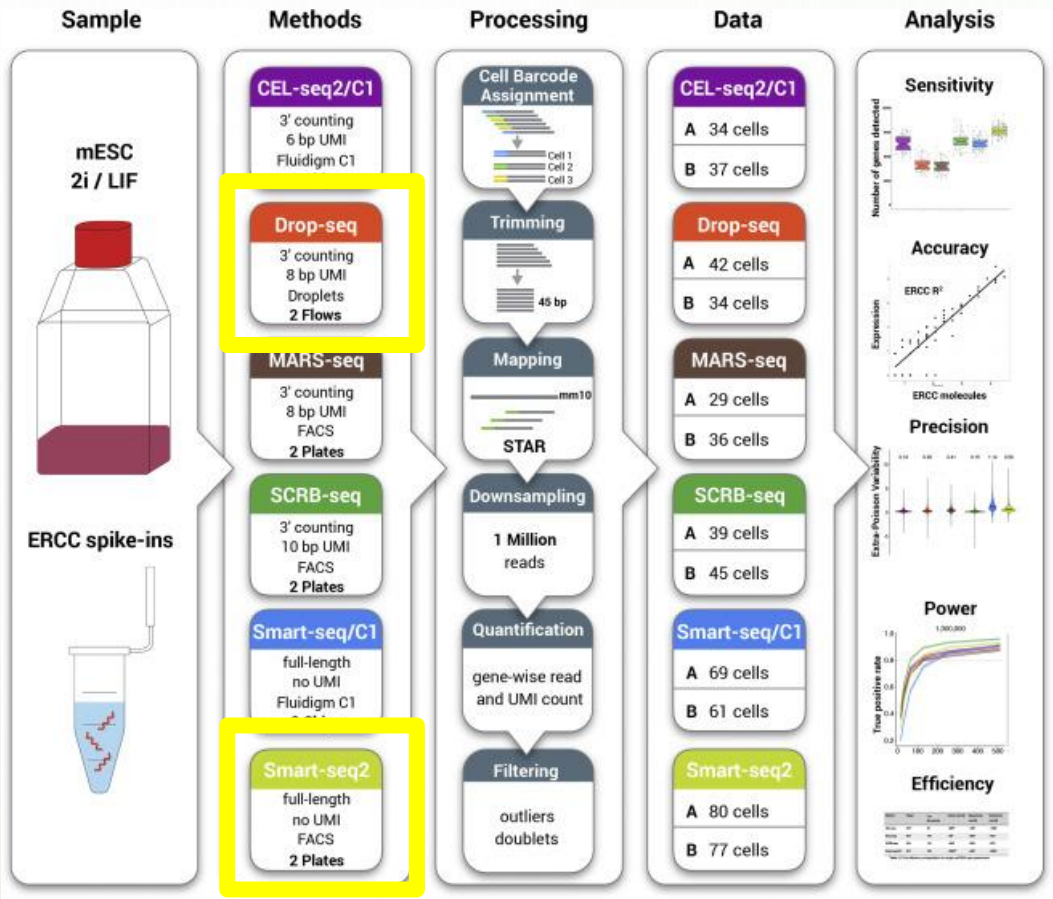
SINGLE CELLS RNA SEQUENCING IN HYPOXIA



NN days

NN days

Single-cell RNA sequencing technologies



https://www.sciencedirect.com/science/article/pii/S1097276517300497

https://emea.illumina.com/science/sequencing-method-explorer/kits-and-arrays/drop-seq.html
https://emea.illumina.com/science/sequencing-method-explorer/kits-and-arrays/smart-seq-nanocage-cagescan.html

Datasets Schema

Technologies

Cell Lines	SMARTseq	DROPseq
MCF7		
HCC1806		

You have an **"unfiltered"** data file. These are data are matrixes of counts, no filter was applied and no normalization was applied. It is a good idea to start to explore this first to understand the issues with the dataset and discuss them, and what are the general implications of the issues you encounter.

You have also **"filtered"** data and **"normalized"** data. These have been pre-processed for you so eliminate some of the issues. You can use these to train your classifiers.